Optimization Methods in Machine Learning, CUB, Spring 2026

Theory 3. Constrained optimization and dual problems.

Task 1 Solve the following optimization problem:

$$\min_{x\in\mathbb{R}^n}\{\langle c,x\rangle : \langle Ax,x\rangle \le 1\},$$

where $c \in \mathbb{R}^n\backslash\{0\}$ and $A \in \mathbb{S}^n_{++}$.

Task 2 Solve the following optimization problem:

$$\min_{X\in\mathbb{S}^n_{++}}\left\{\mathrm{tr}(X^{-1}) : \langle A,X\rangle \le b\right\},$$

where $A \in \mathbb{S}^n_{++}$, $b > 0$.

Task 3 Suppose that $e_1,\dots,e_n$ – a standard basis in $\mathbb{R}^n$. Show that the following optimization problem has a unique solution equal to I_n:

$$\max_{X\in\mathbb{S}^n_{++}}\{\det(X) : \|Xe_i\| \le 1 \text{ for all } 1 \le i \le n\}$$

Derive from here a particular case of Hadamard's inequality:

$$\det(X) \le \|Xe_1\|\dots\|Xe_n\|,$$

that is valid for arbitrary matrix $X \in \mathbb{S}^n_{++}$.

Task 4 Show that the following optimization problem has a unique solution and find it:

$$\min_{X\in\mathbb{S}^n_{++}}\left\{\langle C^{-1},X\rangle - \log\det(X) : \langle Xa,a\rangle \le 1\right\},$$

where $C \in \mathbb{S}^n_{++}$, $a \in \mathbb{R}^n$, $a \neq 0$. The answer should not depend on inverse matrix C^{-1} (hint: use Woodbury identity).

Task 5 Construct the dual optimization problem for the following primal one:

$$\min_{x\in\mathbb{R}^n, s\in\mathbb{R}^m}\left\{\frac{1}{2}\|s-b\|^2 + \frac{\rho}{2}\|x\|^2 : s = Ax\right\},$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $\rho > 0$.

Task 6 Let's consider SVM for K classes (Crammer-Singer formulation). Here the decision rule is

$$y(\boldsymbol{x}) = \arg\max_{k\in\{1,\dots,K\}}(\boldsymbol{w}_k^T\boldsymbol{x} + b_k).$$

The primal QP problem is

$$\frac{1}{2}\sum_{k=1}^{K}\|\boldsymbol{w}_k\|^2 + C\sum_{i=1}^{N}\xi_i \to \min_{W,\boldsymbol{b},\boldsymbol{\xi}},$$
$$\boldsymbol{w}_{y_i}^T\boldsymbol{x}_i + b_{y_i} \ge \boldsymbol{w}_k^T\boldsymbol{x}_i + b_k + 1 - \xi_i,\ \forall k \neq y_i, \forall i,$$
$$\xi_i \ge 0\ \forall i$$

Two inequalities here can be united as one inequality:

$$\boldsymbol{w}_{y_i}^T\boldsymbol{x}_i + b_{y_i} \ge \boldsymbol{w}_k^T\boldsymbol{x}_i + b_k + [k \neq y_i] - \xi_i\ \forall k$$

Show that the dual optimization problem can be expressed as follows:

$$\sum_{i,k=1}^{N,K}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{N}\boldsymbol{x}_i^T\boldsymbol{x}_j\sum_{k=1}^{K}\alpha_{ik}\alpha_{jk} \to \max_{\alpha},$$
$$\sum_{k=1}^{K}\alpha_{ik} = 0\ \forall i = 1,\dots,N;$$
$$\alpha_{ik} \le 0\ \forall i, k \neq y_i, \alpha_{iy_i} \le C.$$